

## SEGEN-KYUEL', Russian Federation

WMO: 246610

Lat: 64.000N

Lon: 130.300E

Elev: 684

StdP: 14.34

Time Zone: 9.00 (E09)

Period: 09-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |             |             |            | MCWS/PCWD<br>to 99.6% DB |            | WSF        |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|-------------|-------------|------------|--------------------------|------------|------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |             | 1%          |            |                          |            |            |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB        | WS          | MCDB       | MCWS                     | PCWD       |            |
| (a)<br>1      | (b)<br>-58.9 | (c)<br>-56.5 | (d)<br>-61.9                  | (e)<br>0.1 | (f)<br>-57.8 | (g)<br>-59.5 | (h)<br>0.2 | (i)<br>-56.3 | (j)<br>22.7           | (k)<br>-3.1 | (l)<br>20.6 | (m)<br>2.1 | (n)<br>0.2               | (o)<br>N/A | (p)<br>N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 22.4                   | 84.2            | 65.4 | 80.8 | 64.0 | 77.3 | 62.1 | 68.2                | 79.6 | 66.0 | 76.6 | 63.9 | 74.0 | 4.2                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 64.1                            | 91.9 | 73.1 | 62.1 | 85.5 | 70.6 | 59.9 | 78.9 | 68.3 | 33.0          | 79.0 | 31.2 | 76.5 | 29.6 | 74.0 | 75.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |           |                    |         | n-Year Return Period Values of Extreme Temperature |           |            |           |            |           |            |           |          |  |
|-------------------|----------|----------|----------------------------|-----------|--------------------|---------|----------------------------------------------------|-----------|------------|-----------|------------|-----------|------------|-----------|----------|--|
|                   |          |          | Mean                       |           | Standard Deviation |         | n=5 years                                          |           | n=10 years |           | n=20 years |           | n=50 years |           |          |  |
| 1%                | 2.5%     | 5%       | Min                        | Max       | Min                | Max     | Min                                                | Max       | Min        | Max       | Min        | Max       | Min        | Max       |          |  |
| (a) 18.5          | (b) 15.7 | (c) 13.1 | (e) -61.4                  | (f) 90.1  | (g) 3.2            | (h) 2.9 | (i) -63.7                                          | (j) 92.2  | (k) -65.5  | (l) 93.9  | (m) -67.3  | (n) 95.6  | (o) -69.6  | (p) 97.7  |          |  |
|                   |          |          | DB                         |           |                    |         |                                                    |           |            |           |            |           |            |           |          |  |
|                   |          |          | WB                         |           |                    |         |                                                    |           |            |           |            |           |            |           |          |  |
|                   |          |          |                            | (e) -61.4 | (f) 72.0           | (g) 3.2 | (h) 2.7                                            | (i) -63.7 | (j) 73.9   | (k) -65.5 | (l) 75.5   | (m) -67.3 | (n) 77.0   | (o) -69.6 | (p) 78.9 |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 14.5   | -39.5 | -30.3 | -3.2  | 24.5  | 43.9  | 59.2  | 63.8  | 57.5  | 41.3  | 14.6  | -21.7 | -38.8 |
|                                                                        | DBStd       | 39.50  | 12.02 | 10.96 | 14.57 | 11.28 | 7.95  | 6.88  | 5.94  | 6.01  | 7.73  | 13.30 | 14.80 | 11.93 |
|                                                                        | HDD50       | 13948  | 2774  | 2247  | 1649  | 764   | 221   | 8     | 1     | 11    | 275   | 1096  | 2150  | 2752  |
|                                                                        | HDD65       | 18508  | 3239  | 2667  | 2114  | 1214  | 654   | 196   | 96    | 239   | 711   | 1561  | 2600  | 3217  |
|                                                                        | CDD50       | 1004   | 0     | 0     | 0     | 0     | 32    | 284   | 429   | 245   | 14    | 0     | 0     | 0     |
|                                                                        | CDD65       | 89     | 0     | 0     | 0     | 0     | 0     | 23    | 59    | 7     | 0     | 0     | 0     | 0     |
|                                                                        | CDH74       | 1514   | 0     | 0     | 0     | 0     | 5     | 452   | 903   | 154   | 0     | 0     | 0     | 0     |
|                                                                        | CDH80       | 369    | 0     | 0     | 0     | 0     | 0     | 97    | 254   | 18    | 0     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 3.5    | 2.1   | 2.5   | 4.0   | 5.2   | 4.8   | 3.8   | 4.1   | 3.7   | 4.4   | 4.1   | 2.1   | 1.5   |
| (15) Precipitation                                                     | PrecAvg     | 16.0   | 0.4   | 0.3   | 0.4   | 0.6   | 1.6   | 2.2   | 3.0   | 2.4   | 2.4   | 1.7   | 0.7   | 0.3   |
|                                                                        | PrecMax     | 21.8   | 0.9   | 0.9   | 2.3   | 1.9   | 3.4   | 4.2   | 6.9   | 5.0   | 5.0   | 3.4   | 1.6   | 1.2   |
|                                                                        | PrecMin     | 9.8    | 0.1   | 0.0   | 0.0   | 0.1   | 0.0   | 0.6   | 0.9   | 0.1   | 0.0   | 0.1   | 0.1   | 0.0   |
|                                                                        | PrecStd     | 2.9    | 0.2   | 0.2   | 0.5   | 0.4   | 0.9   | 0.9   | 1.6   | 1.3   | 1.2   | 0.9   | 0.4   | 0.2   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 13.2  | 4.5   | 33.0  | 51.5  | 73.1  | 86.6  | 90.8  | 82.6  | 69.9  | 42.3  | 23.5  | 12.4  |
|                                                                        |             | MCWB   | 10.1  | 3.1   | 27.0  | 39.4  | 53.1  | 65.5  | 69.2  | 64.6  | 57.2  | 35.6  | 21.6  | 9.0   |
|                                                                        | 2%          | DB     | 1.4   | 0.7   | 28.4  | 47.4  | 67.8  | 82.9  | 85.9  | 78.2  | 63.2  | 38.2  | 12.6  | -5.6  |
|                                                                        |             | MCWB   | -0.5  | -0.7  | 24.0  | 37.6  | 50.8  | 62.2  | 67.3  | 62.7  | 51.3  | 32.1  | 11.2  | -6.4  |
|                                                                        | 5%          | DB     | -8.5  | -3.4  | 24.1  | 43.7  | 62.8  | 79.4  | 82.5  | 74.7  | 58.7  | 34.7  | 8.1   | -12.2 |
|                                                                        |             | MCWB   | -9.4  | -4.6  | 20.3  | 34.9  | 48.6  | 61.5  | 65.7  | 61.1  | 48.6  | 29.0  | 6.7   | -12.6 |
|                                                                        | 10%         | DB     | -22.8 | -10.6 | 20.2  | 40.6  | 58.7  | 75.6  | 79.2  | 71.0  | 54.5  | 31.7  | 2.9   | -21.3 |
|                                                                        |             | MCWB   | -23.1 | -11.2 | 16.9  | 32.7  | 46.8  | 60.2  | 64.5  | 59.1  | 46.3  | 27.8  | 1.9   | -21.5 |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 10.3  | 3.2   | 27.7  | 40.1  | 54.3  | 67.7  | 72.5  | 68.0  | 56.7  | 39.2  | 21.9  | 9.3   |
|                                                                        |             | MCDB   | 13.4  | 4.7   | 31.9  | 50.2  | 70.6  | 80.2  | 85.7  | 78.0  | 68.0  | 40.4  | 22.6  | 12.6  |
|                                                                        | 2%          | WB     | -0.6  | -0.8  | 24.3  | 37.9  | 51.7  | 65.1  | 69.9  | 65.5  | 53.2  | 32.4  | 11.5  | -6.1  |
|                                                                        |             | MCDB   | 1.4   | 0.6   | 28.3  | 47.0  | 65.0  | 78.3  | 82.6  | 74.2  | 59.6  | 35.9  | 12.6  | -5.4  |
|                                                                        | 5%          | WB     | -9.6  | -4.5  | 20.5  | 35.6  | 49.8  | 63.0  | 67.7  | 63.2  | 50.5  | 30.0  | 6.6   | -12.6 |
|                                                                        |             | MCDB   | -8.6  | -3.4  | 23.9  | 42.1  | 60.6  | 75.7  | 79.0  | 71.3  | 56.8  | 34.1  | 8.0   | -12.1 |
|                                                                        | 10%         | WB     | -23.1 | -11.3 | 17.0  | 33.5  | 47.5  | 61.2  | 65.6  | 60.7  | 47.4  | 27.7  | 2.1   | -21.4 |
|                                                                        |             | MCDB   | -22.8 | -10.7 | 20.4  | 39.6  | 57.1  | 73.4  | 75.9  | 67.2  | 53.1  | 31.3  | 3.1   | -21.3 |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 14.0  | 23.6  | 27.5  | 19.9  | 18.2  | 22.7  | 22.4  | 21.7  | 17.3  | 16.4  | 16.4  | 11.9  |
|                                                                        | 5% DB       | MCDBR  | 25.5  | 25.0  | 26.3  | 18.8  | 26.2  | 31.1  | 31.7  | 29.6  | 25.5  | 15.5  | 17.5  | 23.6  |
|                                                                        |             | MCWBR  | 24.3  | 23.8  | 22.2  | 11.9  | 13.4  | 14.3  | 15.3  | 15.5  | 15.6  | 11.3  | 16.5  | 22.8  |
|                                                                        | 5% WB       | MCDBR  | 25.3  | 24.6  | 25.9  | 16.4  | 23.4  | 27.6  | 27.9  | 25.4  | 23.6  | 13.7  | 16.8  | 23.6  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.180 | 0.240 | 0.289 | 0.341 | 0.400 | 0.393 | 0.477 | 0.446 | 0.330 | 0.258 | 0.183 | 0.170 |
|                                                                        | taud        |        | 1.928 | 2.110 | 2.087 | 2.036 | 2.120 | 2.324 | 2.062 | 2.104 | 2.418 | 2.254 | 1.924 | 1.896 |
|                                                                        | Ebn at Noon |        | 131   | 218   | 250   | 260   | 254   | 261   | 230   | 224   | 240   | 209   | 121   | 57    |
|                                                                        | Edh at Noon |        | 9     | 19    | 31    | 41    | 42    | 35    | 45    | 39    | 23    | 16    | 8     | 4     |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 67    | 347   | 928   | 1569  | 1728  | 1820  | 1648  | 1253  | 750   | 353   | 112   | 25    |
|                                                                        | RadStd      |        | 4     | 14    | 46    | 58    | 162   | 125   | 118   | 111   | 92    | 29    | 7     | 1     |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (4 neighbors) | +0.99 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -292  | N/A   | N/A   |

Nomenclature: See separate page